Class 11: Alpha Fold

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Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

This major breakthrough (Figure 1) promises to place Molecular Biology in a new era where we can visualize, analyze and interpret the structures and functions of all proteins.

The PDB Database (main repository of experimental structures) only has ~**250k structures** (saw this in last lab). The main sequence database has over **200 million** sequences! Only 0.125% of known sequences have a known structure - this is called the "structure knowledge gap."

```
(250000/200000000)*100
```

```
[1] 0.125
```

- Structures are much harder to determine than sequences.
- They are expensive (average ~1 million each).
- They take an average of 3-5 years to complete

EBI AlphaFold Database

The EBI has a database of pre-computed alphafold (AF) models called AFDB. This is growing all the time and can be useful to check before running AF ourselves.

Running AlphaFold

We can download and run locally but we need a GPU or we can use "cloud" computing to run this on someone else's computer :D

We will use ColabFold < https://github.com/sokrypton/ColabFold >

We previously found there was no AFDB entry for our HIV sequence:

```
>HIV-Pr-Dimer
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF:PQITLWQRPLVTIKIGGQLK
EALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPT
PVNIIGRNLLTQIGCTLNF
```

Here we will use AlphaFold2_mmseqs2

Custom analysis of resulting models

In this section we will read the results of the more complicated HIV-Pr dimer AlphaFold2 models into R with the help of the Bio3D package.

```
results_dir <- "hivprdimer_23119/"
```

```
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

basename(pdb_files)
```

```
[1] "hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000
hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
.....

Extracting sequences

pdb/seq: 1   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_
pdb/seq: 2   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_
pdb/seq: 3   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_
pdb/seq: 4   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_
pdb/seq: 5   name: hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_
```

```
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]hivprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivprdimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]hivprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivprdimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .         150
[Truncated_Name:1]hivprdimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivprdimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivprdimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivprdimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivprdimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
```

```
                                 **************************************************
                               101        .         .         .         .         150

                               151        .         .         .         .       198
[Truncated_Name:1]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Segment:2]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Segment:3]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Segment:4]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Segment:5]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                                 **************************************************
                               151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD is a standard measure of structural distance between coordinate sets. We are able to use the `rmsd()` function to calculate the RMSD between all pairs models.

```
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

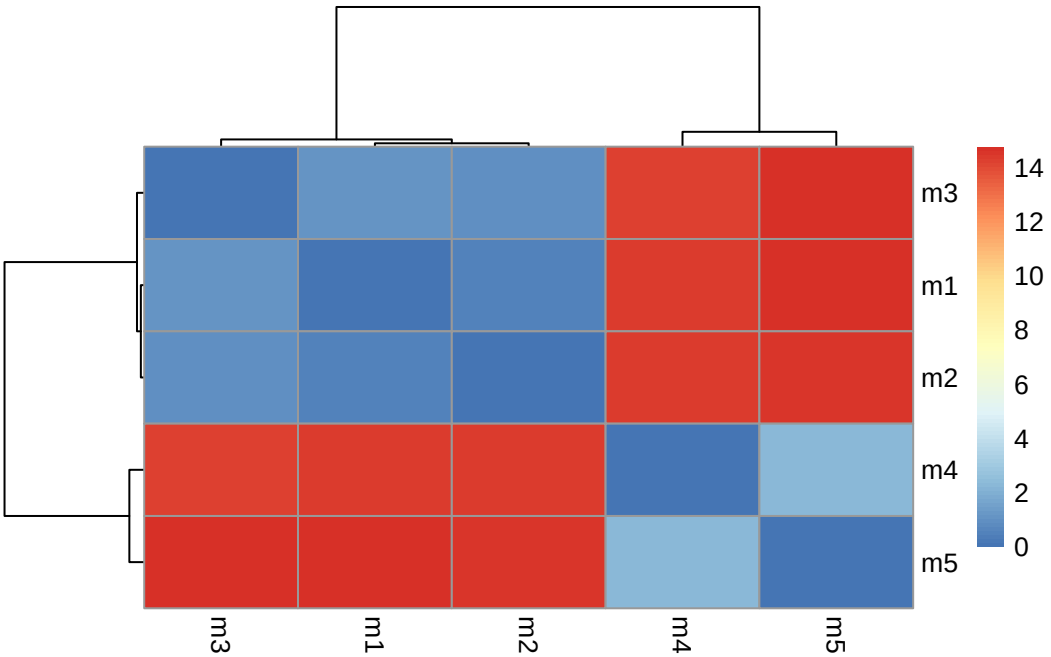
```
range(rd)
```

```
[1]  0.000 14.754
```

Based on these rmsd values we can make a heatmap!

```
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



Now lets plot the pLDDT values across all models. Additionally we will improve the superposition of our models by finding the most consistent "rigid core" common across all the models.

```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
core <- core.find(pdbs)
```

```
 core size 197 of 198  vol = 9836.196
 core size 196 of 198  vol = 6800.793
 core size 195 of 198  vol = 1322.091
 core size 194 of 198  vol = 1032.249
 core size 193 of 198  vol = 944.133
 core size 192 of 198  vol = 894.048
 core size 191 of 198  vol = 829.724
 core size 190 of 198  vol = 766.628
 core size 189 of 198  vol = 728.681
 core size 188 of 198  vol = 693.317
 core size 187 of 198  vol = 656.166
 core size 186 of 198  vol = 622.063
```

```
core size 185 of 198  vol = 586.601
core size 184 of 198  vol = 565.358
core size 183 of 198  vol = 532.747
core size 182 of 198  vol = 511.421
core size 181 of 198  vol = 489.077
core size 180 of 198  vol = 468.599
core size 179 of 198  vol = 449.222
core size 178 of 198  vol = 433.122
core size 177 of 198  vol = 418.651
core size 176 of 198  vol = 405.069
core size 175 of 198  vol = 391.855
core size 174 of 198  vol = 381.006
core size 173 of 198  vol = 371.54
core size 172 of 198  vol = 355.539
core size 171 of 198  vol = 345.156
core size 170 of 198  vol = 335.982
core size 169 of 198  vol = 325.275
core size 168 of 198  vol = 313.603
core size 167 of 198  vol = 302.908
core size 166 of 198  vol = 293.343
core size 165 of 198  vol = 284.547
core size 164 of 198  vol = 277.932
core size 163 of 198  vol = 269.592
core size 162 of 198  vol = 258.167
core size 161 of 198  vol = 246.856
core size 160 of 198  vol = 239.073
core size 159 of 198  vol = 234.174
core size 158 of 198  vol = 226.309
core size 157 of 198  vol = 221.212
core size 156 of 198  vol = 214.951
core size 155 of 198  vol = 206.027
core size 154 of 198  vol = 197.301
core size 153 of 198  vol = 190.889
core size 152 of 198  vol = 185.403
core size 151 of 198  vol = 179.154
core size 150 of 198  vol = 174.709
core size 149 of 198  vol = 167.269
core size 148 of 198  vol = 160.866
core size 147 of 198  vol = 156.173
core size 146 of 198  vol = 148.395
core size 145 of 198  vol = 143.389
core size 144 of 198  vol = 138.428
core size 143 of 198  vol = 132.99
```

```
core size 142 of 198  vol = 126.924
core size 141 of 198  vol = 121.257
core size 140 of 198  vol = 116.407
core size 139 of 198  vol = 112.241
core size 138 of 198  vol = 107.813
core size 137 of 198  vol = 104.807
core size 136 of 198  vol = 100.968
core size 135 of 198  vol = 97.21
core size 134 of 198  vol = 92.54
core size 133 of 198  vol = 87.964
core size 132 of 198  vol = 85.621
core size 131 of 198  vol = 81.514
core size 130 of 198  vol = 77.627
core size 129 of 198  vol = 74.881
core size 128 of 198  vol = 72.642
core size 127 of 198  vol = 70.352
core size 126 of 198  vol = 68.627
core size 125 of 198  vol = 66.349
core size 124 of 198  vol = 64.019
core size 123 of 198  vol = 61.763
core size 122 of 198  vol = 58.758
core size 121 of 198  vol = 56.354
core size 120 of 198  vol = 54.129
core size 119 of 198  vol = 51.611
core size 118 of 198  vol = 49.502
core size 117 of 198  vol = 48.045
core size 116 of 198  vol = 46.489
core size 115 of 198  vol = 44.599
core size 114 of 198  vol = 43.185
core size 113 of 198  vol = 40.978
core size 112 of 198  vol = 39.029
core size 111 of 198  vol = 36.363
core size 110 of 198  vol = 33.953
core size 109 of 198  vol = 31.343
core size 108 of 198  vol = 29.304
core size 107 of 198  vol = 27.188
core size 106 of 198  vol = 25.655
core size 105 of 198  vol = 24.03
core size 104 of 198  vol = 22.504
core size 103 of 198  vol = 20.906
core size 102 of 198  vol = 19.803
core size 101 of 198  vol = 18.215
core size 100 of 198  vol = 15.652
```

```
 core size 99 of 198  vol = 14.762
 core size 98 of 198  vol = 11.574
 core size 97 of 198  vol = 9.391
 core size 96 of 198  vol = 7.327
 core size 95 of 198  vol = 6.123
 core size 94 of 198  vol = 5.592
 core size 93 of 198  vol = 4.689
 core size 92 of 198  vol = 3.66
 core size 91 of 198  vol = 2.779
 core size 90 of 198  vol = 2.147
 core size 89 of 198  vol = 1.725
 core size 88 of 198  vol = 1.158
 core size 87 of 198  vol = 0.88
 core size 86 of 198  vol = 0.682
 core size 85 of 198  vol = 0.527
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

Now the identified core position will be used as a basis for a more suitable superposition in

```r
core.inds <- print(core, vol=0.5)
```

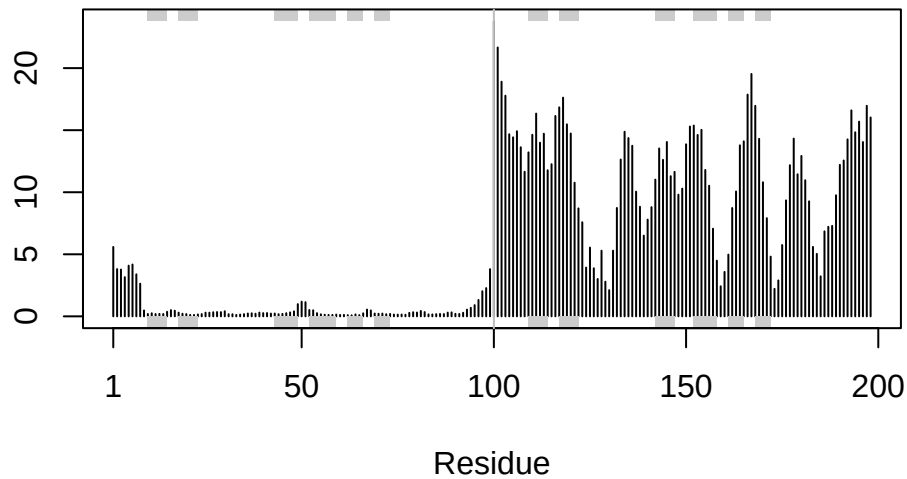
```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

Now we can look at the RMSF between positions in the structure. It is a measure of conformational variance along the structure

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

AlphaFold produces an output called ***Predicted Aligned Error (PAE)**

```
library(jsonlite)

pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

We can also look at the max pae to make ranking models. We can see pae1 is better than 5 bc the lower the PAE score the better.

```
pae1$max_pae
```

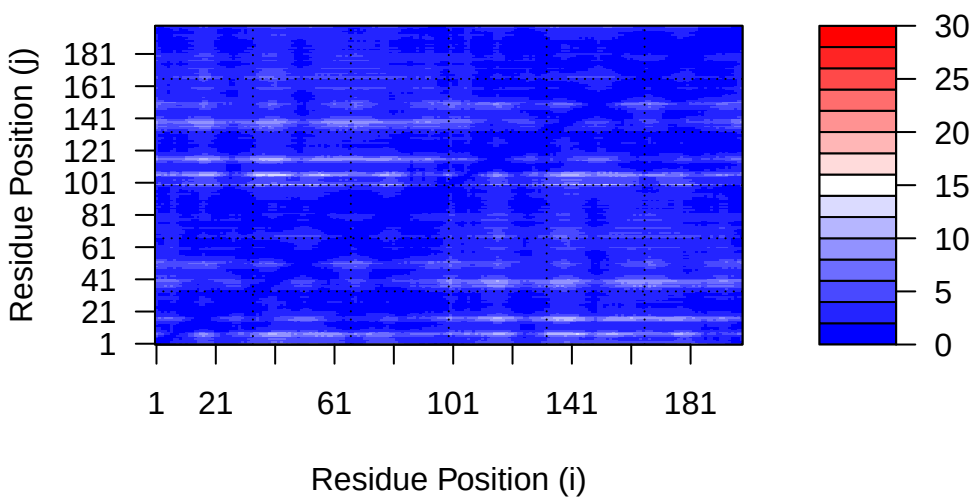
```
[1] 12.84375
```

```
pae5$max_pae
```

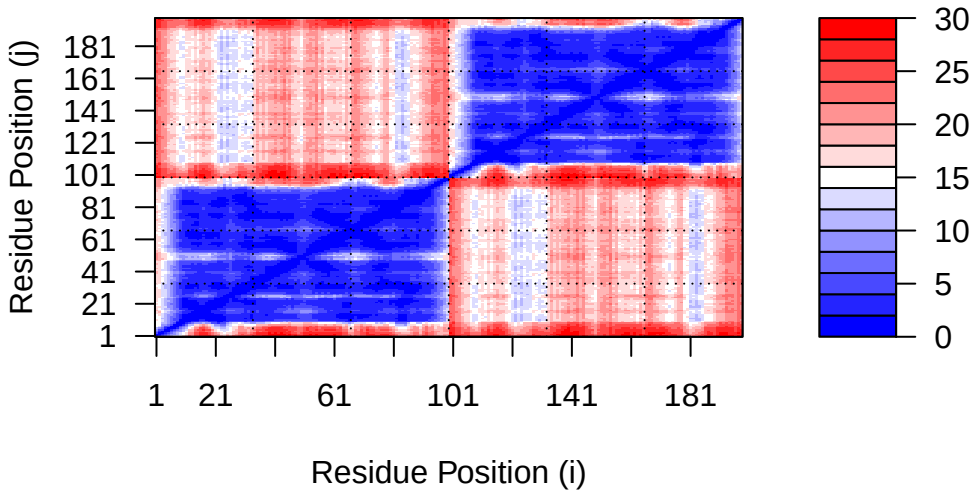
```
[1] 29.59375
```

Lets make a plot of the number of residues by the pae score of number of residues

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivprdimer_23119//hivprdimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

How many sequences are in this alignment

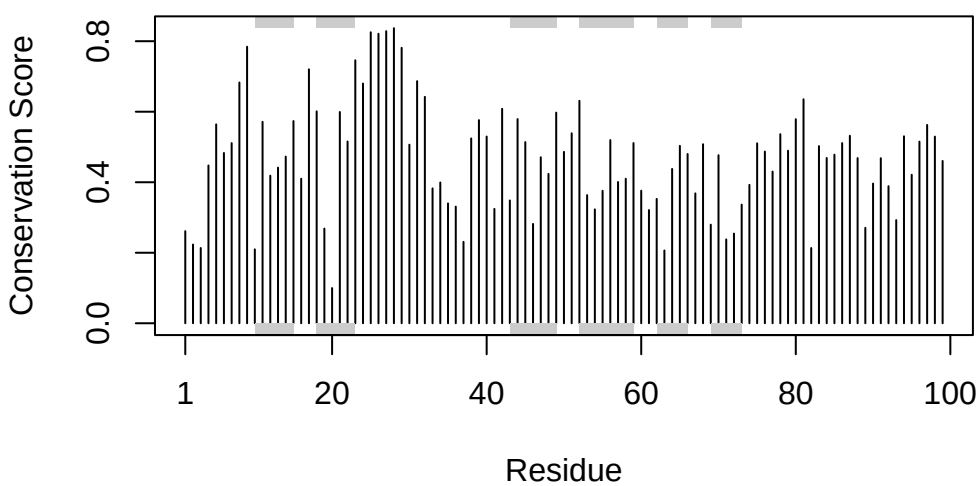
```
dim(aln$ali)
```

```
[1] 5397  132
```

Lets score residue conservation in the alignment with the `conserv()` function

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Conserved active Site residues D25, T26, G27, A28

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

```
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

For a final visualization of these functionally important sites we can map this conservation score to the Occupancy column of a PDB file.

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

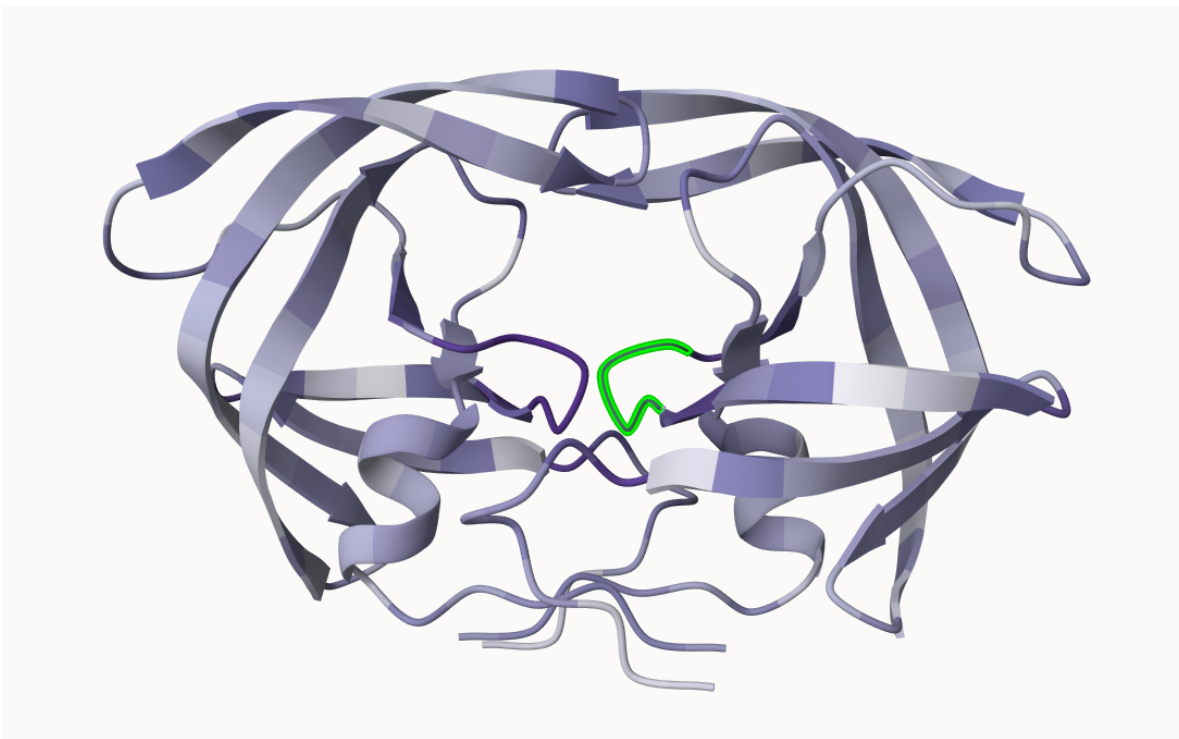


Figure 1: Figure is colored by sequence conservation w/ conserved positions in a darker purple. The DTGA motif is highlighted in green